

Notes: Gopinath and Neiman (AER, 2014)

Trade Adjustment and Productivity in Large Crises

Contents

1	Big picture	2
1.1	Question	2
1.2	Setting	2
1.3	Core finding	2
1.4	Economic takeaway	2
2	Incremental contribution of the paper	3
2.1	Relative to trade-margin papers	3
2.2	Relative to imported-input productivity papers	3
2.3	Relative to terms-of-trade and welfare formulas	3
2.4	Relative to crisis and TFP studies	3
3	Data and measurement	3
3.1	Argentine crisis background	3
3.2	Customs transaction data	4
3.3	Capital IQ firm information	4
3.4	Margin definitions	4
3.5	Sub-extensive and sub-intensive margins	4
4	Empirical findings	5
4.1	Finding 1: Country-level extensive margins are small	5
4.2	Finding 2: Within-firm input churning is large	5
4.3	Finding 3: Adjustment varies systematically with importer size	5
4.4	Finding 4: Dropped input varieties raise firm-level import bundle costs	5
5	Model: Multi-input firms, trade, and productivity	6
5.1	Production technology	6
5.2	Roundabout production	6
5.3	Final-good aggregator	6
5.4	Fixed costs of importing varieties	6
5.5	Unit cost and input price indices	7
5.6	Firm first-order conditions	7

5.7	Domestic input share	7
6	Productivity derivations	8
6.1	Modified Solow residual	8
6.2	Firm-level productivity response	8
6.3	Economic meaning of each term	8
6.4	Sector-level productivity and firm heterogeneity	8
6.5	Measured productivity and missing varieties	9
7	Calibration and simulation results	9
7.1	Baseline calibration	9
7.2	How well the simulation matches the data	9
7.3	Baseline productivity effects	9
7.4	Role of roundabout production	10
7.5	Robustness	10
8	Time-series evidence for Argentina	10
8.1	Import share, dropped varieties, and TFP	10
8.2	Interpretation	10
9	Extracted figures and tables from the paper	11
9.1	Figure 1: Sample Quarterly Product Imports for Two Large Argentine Firms and Figure 2: Our Dataset Compared to Other Sources	11
9.2	Table 1: Import Summary Statistics and Table 2: Argentina’s 50 Largest Importers, Rows 1–23	11
9.3	Table 2: Argentina’s 50 Largest Importers, Rows 24–50 and Figure 3: Number of Importing Firms and Products	12
9.4	Figure 4: Various Extensive Margin Definitions by Quarterly and Annual Data and Table 3: Intensive and Extensive Margins, 2000–2002	13
9.5	Figure 5: Sub-Extensive Margin and Table 4: Sub-Intensive, Sub-Extensive, and Extensive Margins, 2000–2002	14
9.6	Figure 6: Different Firms Drop Different Products and Figure 7: Adjustment by Importer Size, 2000–2002	14
9.7	Table 5: Impact of Product Extensive Margin on Imported Input Costs, 2000–2002 and Figure 8: Inventory Intensity and Import Decline	15
9.8	Figure 9: Export Varieties Did Not Decline Along with Import Varieties and Table 6: Baseline Simulation Parameters	16
9.9	Figure 10: Concentration of Imports in the Data (2000) and in the Simulation and Figure 11: Adjustment and Size in Simulation	16

9.10	Figure 12: Changes in Domestic Input Share and Productivity and Table 7: Simulation Results: Productivity	17
9.11	Figure 13: Time-Series Paths of Productivity, Dropped Varieties, and Import Shares	18
10	Short summary	19
11	Conclusion	19

1 Big picture

1.1 Question

Gopinath and Neiman ask how trade adjusts during a large macroeconomic crisis and why this adjustment matters for productivity. The setting is Argentina’s 2001–2002 crisis, during which the peso depreciated sharply, imports collapsed, and measured total factor productivity fell substantially.

The paper has two linked questions. First, mechanically, did the import collapse occur because firms exited importing, because products disappeared from the country’s import bundle, or because continuing firms changed what and how much they imported? Second, economically, can the collapse in imported intermediate inputs generate a meaningful fall in a welfare-relevant measure of manufacturing productivity?

1.2 Setting

The empirical analysis uses detailed Argentine customs transaction data covering the universe of imports during 1996–2008. The period includes the pre-crisis expansion, the sudden stop and currency crisis, and the recovery. The key episode is the 69 percent drop in dollar imports from 2000 to 2002.

The paper combines the customs data with Capital IQ information on the largest firms. This allows the authors to distinguish changes in imports across firms, across products, across source countries, and across firm-product-country combinations.

1.3 Core finding

The paper’s central empirical claim is that the usual country-level extensive margin is misleadingly small. Firm entry and exit and country-level product entry and exit explain only a small share of the import collapse. By contrast, continuing firms substantially changed the set of imported inputs they used. The authors call this within-firm product churning the **sub-extensive margin**.

The central quantitative claim is that the sub-extensive margin explains up to 45 percent of the 69 percent decline in imports, depending on the product variety definition. This adjustment is invisible in aggregate data because one firm may stop importing an input while another firm continues to import it.

1.4 Economic takeaway

Imports in crises are not just final goods disappearing from the country. Much of the adjustment happens inside firms’ production functions: firms drop some foreign input varieties, reduce quantities of continuing imported inputs, and substitute toward domestic inputs. If imported inputs are imperfect substitutes for domestic ones, losing input varieties raises unit production costs. With markups and roundabout production, these cost increases reduce scale and measured productivity.

2 Incremental contribution of the paper

2.1 Relative to trade-margin papers

Earlier work showed that aggregate country-level extensive margins can be small. This paper shows that this conclusion can be incomplete for intermediate inputs. A product can remain in the country's import bundle while disappearing from many individual firms' input bundles. The relevant extensive margin for production costs is therefore firm-level, not just country-level.

The paper introduces and measures a margin that is between the traditional intensive and extensive margins. It is not firm exit from importing, and it is not total product exit from the country. It is the dropping of imported varieties by continuing importers.

2.2 Relative to imported-input productivity papers

The paper differs from studies of trade liberalization that emphasize productivity gains from access to imported inputs. Here the shock is a large crisis that makes imported inputs more expensive and induces firms to drop input varieties. The analysis therefore studies the reverse channel: a collapse in imported intermediate inputs can lower productivity.

The paper also distinguishes two productivity concepts. A decline in firms' unit costs is not the same as a decline in a welfare-relevant Solow residual. The authors explicitly derive how markups, intermediate input intensity, fixed import costs, and roundabout production translate trade adjustment into productivity changes.

2.3 Relative to terms-of-trade and welfare formulas

In a broad class of trade models, welfare changes can be summarized by aggregate trade shares and one elasticity. This paper shows that such sufficiency fails in their environment. With heterogeneous firms and fixed costs of imported varieties, the full distribution of firm-level import-share changes matters. Two economies can have the same aggregate import share change but different productivity losses because the firm-level pattern of dropped varieties and market-share reallocation differs.

2.4 Relative to crisis and TFP studies

The paper offers a mechanism linking currency crisis, import collapse, and productivity decline. It does not claim to explain all of Argentina's productivity fall. Instead, it provides a disciplined channel: the crisis raises the effective price of imported inputs, firms drop varieties, unit costs rise, production scale falls, and measured productivity declines.

3 Data and measurement

3.1 Argentine crisis background

Argentina experienced a sharp macroeconomic contraction and exchange-rate collapse around 2001–2002. Real GDP and manufacturing productivity fell, while the dollar value of imports collapsed

by 69 percent from 2000 to 2002. Because dollar-denominated import prices were relatively stable, the peso depreciation translated into a large rise in peso import costs.

This is the ideal setting for the paper because the import price shock was large, concentrated, and observable; the customs microdata are detailed enough to identify firms, products, source countries, and quantities.

3.2 Customs transaction data

The customs data record import and export shipping manifests. For imports, they generally contain importer name, declaration date, source country, quantity, weight, price, value, and detailed product codes at least as disaggregated as 10-digit HTS codes. The authors primarily use Datamyne data and cover nearly the entire 1996–2008 import universe, except for a late-1990s period with incomplete reporting.

The paper validates the customs dataset by comparing aggregate imports in the microdata to IMF International Financial Statistics and to US bilateral trade records. The aggregate patterns line up closely.

3.3 Capital IQ firm information

The customs data contain many more firms than Capital IQ, but trade is highly concentrated. Matching to Capital IQ covers firms accounting for roughly 60–70 percent of imports. This matched sample allows the authors to classify large importers by industry, identify multinationals, and show that the largest importers are mostly producers, not merely distributors.

3.4 Margin definitions

The standard decomposition splits import growth into an intensive margin and an extensive margin. Let $v_{i,t}$ be firm i 's total import spending in period t , and let Ψ_t be the set of importers in period t . Aggregate imports are $v_t = \sum_{i \in \Psi_t} v_{i,t}$. The paper decomposes import growth as

$$\frac{\Delta v_t}{v_{t-1}} = \left(\sum_{i \in \Psi_t \cap \Psi_{t-1}} \frac{\Delta v_{i,t}}{v_{t-1}} \right) + \left(\sum_{i \in \Psi_t, i \notin \Psi_{t-1}} \frac{v_{i,t}}{v_{t-1}} - \sum_{i \notin \Psi_t, i \in \Psi_{t-1}} \frac{v_{i,t-1}}{v_{t-1}} \right).$$

The first term is the intensive margin of continuing importers. The second term is the extensive margin from entering and exiting importers.

The same idea can be applied to products instead of firms. The authors consider product definitions using HTS-6, HTS-10, HTS-by-country, and HTS-10-by-country.

3.5 Sub-extensive and sub-intensive margins

Once the analysis moves inside continuing firms, the paper distinguishes:

- **Sub-extensive margin:** a continuing firm starts or stops importing a product variety.
- **Sub-intensive margin:** a continuing firm continues importing a variety but changes the dollar value imported.

The sub-extensive margin is crucial because firms can drop input varieties even when the country continues importing them.

4 Empirical findings

4.1 Finding 1: Country-level extensive margins are small

Although the number of importers fell by more than half and the number of imported product categories declined sharply, the value-weighted contribution of entry and exit is small. From 2000 to 2002, firm exit explains only about 8 percentage points of the 69 percent decline in imports. Product entry and exit explain very little at the country level unless products are defined extremely finely as HTS-10-by-country.

The reason is trade concentration. The largest importers and largest product categories account for most import value and rarely disappear completely. Extensive margin movements are numerous but low-value.

4.2 Finding 2: Within-firm input churning is large

When the same data are viewed at the firm-product level, the product margin becomes important. Continuing firms drop many imported varieties, even when those varieties continue to be imported by other firms. Depending on product definition, the sub-extensive margin explains 18 to 45 percent of the import decline from 2000 to 2002.

This is the paper's most important empirical insight. Aggregate data suggest the product extensive margin is unimportant; firm-level data reveal it is central for firms' production costs.

4.3 Finding 3: Adjustment varies systematically with importer size

Small importers experience larger percentage drops in imports than large importers. For the smallest firms, exit from importing is common. Among continuing importers, smaller firms rely more on sub-extensive adjustment, while larger firms adjust more on the sub-intensive margin.

This size pattern matters for aggregation. If large firms drop fewer varieties and small firms drop more, then aggregate productivity effects depend on how market shares shift across firms and on which firms face the largest unit-cost increases.

4.4 Finding 4: Dropped input varieties raise firm-level import bundle costs

The authors use the Feenstra CES price-index correction to infer the change in the unit cost of imported input bundles caused by lost varieties. If imported varieties are imperfect substitutes, dropping varieties raises the ideal price index of imported inputs.

For a CES aggregator with elasticity 4, aggregate country-level data imply almost no variety correction. But firm-level data imply that the average import bundle cost rose by about 6.4 to 13.4 percent because firms dropped input varieties.

5 Model: Multi-input firms, trade, and productivity

5.1 Production technology

Each domestic manufacturing firm i produces a differentiated variety using capital, production labor, and intermediate inputs:

$$Y_i = A_i \left(K_i^\alpha L_{p,i}^{1-\alpha} \right)^{1-\mu} X_i^\mu.$$

Here A_i is firm-specific technology, K_i capital, $L_{p,i}$ production labor, and X_i the composite intermediate input bundle.

The intermediate bundle combines domestic and imported input bundles:

$$X_i = (Z_i^\rho + M_i^\rho)^{1/\rho}.$$

Domestic and imported bundles are themselves CES aggregates:

$$Z_i = \left(\int z_{ij}^\theta dj \right)^{1/\theta}, \quad M_i = \left(\int_{k \in \Omega_i} m_{ik}^\theta dk \right)^{1/\theta}.$$

The elasticity of substitution across varieties within each bundle is $1/(1 - \theta)$. The elasticity of substitution between domestic and foreign bundles is $1/(1 - \rho)$.

5.2 Roundabout production

Each firm's output can be used as final consumption or as an intermediate input by other domestic firms:

$$Y_i = g_i + z_i = g_i + \int z_{ji} dj.$$

This creates roundabout production. If an import shock raises one firm's cost, that firm's price rises, which raises the cost of using that firm's output as an intermediate input for other firms. This is the model's amplification mechanism.

5.3 Final-good aggregator

Aggregate manufacturing consumption is a CES aggregate of individual domestic final goods:

$$G = \left(\int g_i^\theta di \right)^{1/\theta}.$$

The same parameter θ governs substitution across domestic varieties in final and intermediate use.

5.4 Fixed costs of importing varieties

A firm pays a labor fixed cost for importing a measure $|\Omega_i|$ of input varieties:

$$F(|\Omega_i|) = f|\Omega_i|^\lambda.$$

This fixed-cost structure makes import demand non-homothetic. Large firms have enough scale to cover the cost of importing more varieties; small firms import fewer varieties and drop more varieties after a shock.

5.5 Unit cost and input price indices

The firm's unit cost is

$$C_i = \frac{1}{A_i} \frac{P_V^{1-\mu} P_{X_i}^\mu}{\mu^\mu (1-\mu)^{1-\mu}},$$

where P_V is the unit cost of the capital-labor bundle and

$$P_{X_i} = \left(P_Z^{\rho/(\rho-1)} + P_{M_i}^{\rho/(\rho-1)} \right)^{(\rho-1)/\rho}.$$

The imported input price index is

$$P_{M_i} = p_m |\Omega_i|^{(\theta-1)/\theta}.$$

Because $\theta < 1$, a larger measure of imported varieties lowers P_{M_i} . Dropping imported input varieties raises the import bundle cost even if the observed price p_m of each surviving variety is unchanged.

5.6 Firm first-order conditions

The cost-minimizing choices imply

$$\begin{aligned} wL_{p,i} &= (1-\mu)(1-\alpha)C_iY_i, \\ rK_i &= (1-\mu)\alpha C_iY_i, \\ P_{X_i}X_i &= \mu C_iY_i, \\ z_{ij} &= \left(\frac{p_j}{P_Z} \right)^{1/(\theta-1)} \left(\frac{P_Z}{P_{X_i}} \right)^{1/(\rho-1)} X_i, \\ m_i &= \left(\frac{p_m}{P_{M_i}} \right)^{1/(\theta-1)} \left(\frac{P_{M_i}}{P_{X_i}} \right)^{1/(\rho-1)} X_i. \end{aligned}$$

The firm sets price as a constant markup over cost,

$$p_i = \frac{C_i}{\theta}.$$

5.7 Domestic input share

Define

$$\gamma_i = \frac{P_Z Z_i}{P_{X_i} X_i},$$

the domestic share in intermediate input spending. Firms with better technology import more varieties, have lower imported-input price indices, and have lower γ_i . A crisis that raises import costs increases γ_i as firms shift toward domestic inputs.

6 Productivity derivations

6.1 Modified Solow residual

The firm-level Solow residual is

$$\Delta \ln PR_i = \Delta \ln Y_i^{VA} - s_{L_i} \Delta \ln L_i - s_{K_i} \Delta \ln K_i,$$

where Y_i^{VA} is value added and s_{L_i} , s_{K_i} are value-added factor shares. At the sector level,

$$\Delta \ln PR = \Delta \ln Y^{VA} - s_L \Delta \ln L - s_K \Delta \ln K.$$

This is the welfare-relevant productivity measure emphasized by Basu and Fernald.

6.2 Firm-level productivity response

The paper derives a first-order approximation:

$$\Delta \ln PR_i = \frac{1 - \theta}{\theta(1 - \mu)} \left[\Delta \ln V_i + \frac{\mu\theta}{1 - \mu\theta} (\Delta \ln X_i - \Delta \ln Y_i) \right] - \frac{1 - \mu\theta}{\theta(1 - \mu)} s_{L_i} (1 - \omega_{L_p, i}) \Delta \ln L_{f, i} + \Delta \ln A_i / (1 - \mu).$$

Here V_i is the capital-labor input bundle, $L_{f, i}$ is labor used to pay fixed import costs, and $\omega_{L_p, i} = L_{p, i} / L_i$.

6.3 Economic meaning of each term

The expression has four parts:

- **Scale effect.** With markups, a decline in scale lowers welfare-relevant productivity even if technology is unchanged. This is the Hall/Basu-Fernald mechanism.
- **Intermediate-input intensity.** If firms use fewer intermediates relative to output, markups make this change first-order relevant for productivity.
- **Fixed-cost labor.** Dropping imported varieties reduces fixed-cost labor, which mechanically raises measured productivity relative to the case where fixed costs remain unchanged.
- **Technology.** The simulations hold A_i fixed, so productivity changes arise through trade and scale mechanisms, not exogenous technology shocks.

6.4 Sector-level productivity and firm heterogeneity

The sectoral response to an import price shock can be decomposed into a scale term, reallocation and non-homothetic demand terms, distributional terms involving firm-level domestic input shares, and the direct import-price term.

The essential point is that with fixed importing costs, γ_i changes differently across firms. Therefore aggregate productivity cannot be summarized only by the aggregate import share. The entire distribution of $\{\gamma_i, \omega_i\}$, including how market shares ω_i shift after the shock, matters.

6.5 Measured productivity and missing varieties

The paper distinguishes true productivity from measured productivity. Statistical agencies typically use matched-model price indices, which follow continuing varieties but do not adjust for disappearing varieties. If firms drop imported varieties and the production function is CES, the true import price index rises more than the measured index.

Thus, if a matched-model price index is used, real intermediate input use is overstated and measured productivity is understated. The paper writes measured productivity as

$$\Delta \ln \widetilde{PR} = \Delta \ln PR - \frac{\mu}{1-\mu} \frac{1-\theta}{1-\mu\theta} \sum_i \omega_i \Delta \ln(P_{X_i}/\widetilde{P}_{X_i}).$$

This is why the sub-extensive margin matters both economically and statistically.

7 Calibration and simulation results

7.1 Baseline calibration

The benchmark simulation uses a manufacturing sector with 1,000 firms. The authors set $\theta = \rho = 0.75$, corresponding to an elasticity of substitution of 4. They set $\mu = 2/3$, consistent with Argentina's input-output table, and choose other parameters to match the import-share distribution and the observed decline in imported input use.

The pre-crisis domestic input share is calibrated to $\gamma^{pre} = 0.83$, and the post-crisis value is $\gamma^{post} = 0.89$. The foreign input price shock is chosen to generate this movement.

7.2 How well the simulation matches the data

The simulated economy replicates the main empirical facts:

- The firm extensive margin is absent by construction and unimportant in the data.
- The sub-extensive margin contributes 47 percent of the simulated decline, close to the 45 percent in the data using the most disaggregated definition.
- Larger simulated firms experience smaller percentage declines in imports and rely more on sub-intensive adjustment.
- The import bundle price correction from dropped varieties is 8.9 percent, in the empirical range of 6–13 percent.

7.3 Baseline productivity effects

In the benchmark model with fixed costs and roundabout production, the import cost shock produces a productivity decline of 5.5 percent. Taking into account mismeasurement from dropped varieties increases the measured decline to 6.6 percent.

The model without fixed costs, and therefore without an active sub-extensive margin, generates a productivity decline of 4.3 percent. Thus the sub-extensive margin makes the productivity decline about 1.2 percentage points larger, roughly a 30 percent increase relative to the no-fixed-cost model.

7.4 Role of roundabout production

Without roundabout production, productivity declines by only 2.6 percent. This is roughly half the benchmark decline. Roundabout production amplifies the import shock because higher costs for one firm raise input costs for other firms.

7.5 Robustness

The authors vary substitution elasticities and consider alternative interpretations of inventories and capital goods. The productivity decline remains economically meaningful across specifications. For example, with a lower substitution elasticity between domestic and foreign inputs, the productivity decline can approach 7 percent; adjusting for inventories reduces but does not eliminate the effect.

8 Time-series evidence for Argentina

8.1 Import share, dropped varieties, and TFP

The paper compares the model’s mechanisms with independent Argentine TFP measures. The share of input spending on imports falls during the crisis and begins recovering afterward. The estimated variety effect also worsens during the crisis and recovers later. These paths line up with the decline and recovery of TFP.

This time-series evidence is not a causal proof by itself, because Argentina experienced many shocks. But it corroborates the paper’s mechanism: the decline and recovery of imported input use and variety effects move in the same direction as productivity.

8.2 Interpretation

The evidence suggests that imported intermediate input use is a meaningful contributor to crisis productivity dynamics. The mechanism is not the only force in Argentina’s crisis, but it is a quantitatively important and previously undermeasured channel.

9 Extracted figures and tables from the paper

9.1 Figure 1: Sample Quarterly Product Imports for Two Large Argentine Firms and Figure 2: Our Dataset Compared to Other Sources

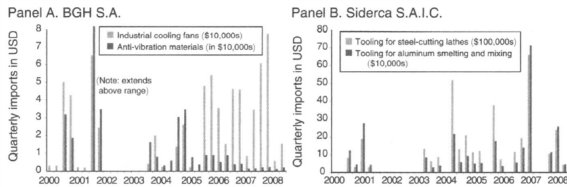


FIGURE 1. SAMPLE QUARTERLY PRODUCT IMPORTS FOR TWO LARGE ARGENTINE FIRMS

Figure 1: Sample quarterly product imports for BGH S.A. and Siderca S.A.I.C.

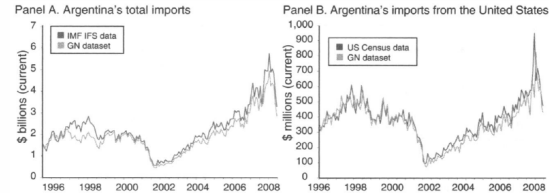


FIGURE 2. OUR DATASET COMPARED TO OTHER SOURCES

Figure 2: Dataset compared to other sources.

Read: Figure 1 shows two large firms dropping specific imported inputs during the crisis while continuing to operate as importers. Figure 2 validates the customs dataset by comparing the authors' data to external aggregate sources.

Economic message: The key margin is not necessarily whether a firm imports at all; it is whether a firm keeps importing the same production inputs. The data validation figure also makes the micro evidence credible for aggregate crisis dynamics.

9.2 Table 1: Import Summary Statistics and Table 2: Argentina's 50 Largest Importers, Rows 1–23

TABLE 1—IMPORT SUMMARY STATISTICS

	2000	2002
Number of importing firms	25,138	13,980
Number of supplier countries		
Economy-wide	135	130
Per firm, median	2	1
Per firm, maximum	44	36
Number of imported HTS codes		
Economy-wide	17,333	15,831
Per firm, median	5	4
Per firm, maximum	899	733
Number of imported country × HTS combinations		
Economy-wide	115,724	80,781
Per firm, median	6	4
Per firm, maximum	2,067	2,176

Table 1: Import summary statistics for 2000 and 2002.

TABLE 2—ARGENTINA'S 50 LARGEST IMPORTERS

	Importer name	Primary industry	Primary sector	Avg. annual imports (\$millions)	Share of imports, 2000	Share of imports, 2002
1	Volkswagen Arg.	Auto mfg.	Cons. disc.	544.5	1.6	0.9
2	Ford Arg.	Auto mfg.	Cons. disc.	508.2	2.0	2.1
3	General Motors de Arg.	Auto mfg.	Cons. disc.	375.7	1.2	2.3
4	Renault Arg.	Auto mfg.	Cons. disc.	368.4	2.3	1.0
5	Peugeot Citroen Arg.	Auto mfg.	Cons. disc.	308.6	1.5	1.0
6	Daimler Chrysler Arg.	Auto mfg.	Cons. disc.	287.3	1.4	0.2
7	Siderar	Steel	Materials	288.8	0.8	1.9
8	Fiat Auto Arg.	Auto mfg.	Cons. disc.	242.1	0.6	0.5
9	YPF	Int. oil & gas	Energy	236.1	0.4	0.8
10	C.T.I.	Wireless telecom.	Telecom. svcs.	210.1	0.2	0.1
11	Telefonica Comm. Per.	Integrated telecom.	Telecom. svcs.	189.0	1.2	0.0
12	Monsanto Arg.	Agr. chems.	Cons. stpls.	176.5	0.9	2.0
13	Hewlett-Packard Arg.	Tech. distrib.	IT	172.1	0.9	0.3
14	Toyota Arg.	Auto mfg.	Cons. disc.	169.6	0.6	0.7
15	Telecom Personal	Wireless telecom.	Telecom. svcs.	165.6	0.8	0.0
16	Petrobras Energia	Int. Oil & gas	Energy	156.4	0.1	0.5
17	Aluar Aluminio Arg.	Aluminum	Materials	138.4	0.6	1.0
18	Acindar Ind. Arg. de Aceros	Steel	Materials	129.9	0.3	0.9
19	Shell Co. Arg. De Petroleo	Oil & gas explo.	Energy	120.6	0.7	1.0
20	Industrias John Deere Arg.	Const. & farm mach.	Industrials	112.1	0.2	0.2
21	Eso Petrolera Arg.	Oil & gas explo.	Energy	103.5	1.1	0.5
22	Siderca	Building products	Industrials	102.3	0.3	0.6
23	Bayer Arg.	Pharmaceuticals	Healthcare	97.4	0.4	0.8

Table 2: Argentina's largest importers, rows 1–23.

Read: Table 1 shows that the number of importers and product-country combinations fell dramatically from 2000 to 2002. Table 2 shows the concentration of import value among a small set of large firms, especially automobile and industrial firms.

Economic message: The count margins look large, but value is concentrated among firms that generally remain importers. This is why value-weighted extensive margins are small.

9.3 Table 2: Argentina's 50 Largest Importers, Rows 24–50 and Figure 3: Number of Importing Firms and Products

24	Honda Motor De Arg.	Auto mfg.	Cons. disc.	92.4	0.2	0.1
25	BGH	Household apps.	Cons. disc.	92.0	0.4	0.2
26	Siemens Arg.	Elect. equip. & inst.	IT	91.6	0.6	0.2
27	Scania Arg.	Auto mfg.	Cons. disc.	91.5	0.4	0.3
28	Productos Roche	Pharmaceuticals	Healthcare	89.5	0.5	0.8
29	Atanor	Comm. chems.	Materials	88.9	0.2	0.4
30	Co. de Radiocom. Mobil.	Wireless telecom.	Telecom. svcs.	87.2	1.1	0.1
31	IBM Arg.	IT consulting	IT	83.7	0.5	0.4
32	Syngenta Agro	Agr. chems.	Materials	82.7	0.1	0.4
33	Alta Plastica	Distributors	Cons. disc.	76.8	0.4	0.3
34	Iveco Arg.	Auto mfg.	Cons. disc.	75.7	0.2	0.1
35	BASF Arg.	Commodity chems.	Materials	73.9	0.3	0.5
36	Pirelli Neumaticos	Tires & rubber	Cons. disc.	73.7	0.2	0.6
37	Minera Alumbrera Limited	Gold	Materials	73.6	0.2	0.8
38	Unilever De Arg.	Household prods.	Cons. disc.	72.9	0.3	0.6
39	Tetra Pak	Pkgd. foods/meats	Cons. stpls.	70.7	0.4	0.7
40	Novartis Arg.	Pharmaceuticals	Healthcare	67.5	0.4	0.4
41	Philips Arg.	Tech. distrib.	IT	66.7	0.3	0.2
42	Procter & Gamble Arg.	Household prods.	Cons. disc.	67.6	0.2	0.3
43	Abbott Laboratories Arg.	Pharmaceuticals	Healthcare	65.8	0.4	0.7
44	Vordian Arg.	Commod. chems.	Materials	65.0	0.2	0.8
45	Bridgestone Firestone Arg.	Auto parts & equip.	Cons. disc.	62.0	0.2	0.4
46	Nidera	Food distributors	Cons. stpls.	63.2	0.2	0.3
47	AGCO Arg.	Const. & farm mach.	Industrials	61.8	0.1	0.1
48	Sipar Aceros	Steel	Materials	60.8	0.1	0.2
49	Aerolineas Arg.	Airlines	Industrials	60.3	0.7	0.2
50	Dow Quimica Arg.	Comm. chems.	Materials	59.1	0.2	0.4

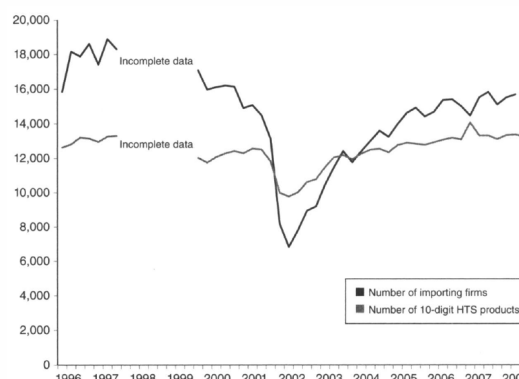


FIGURE 3. NUMBER OF IMPORTING FIRMS AND PRODUCTS

Table 2: Argentina's largest importers, rows 24–50. Figure 3: Number of importing firms and products.

Read: The bottom half of Table 2 continues to show large real-sector firms among the major importers. Figure 3 shows the crisis-period collapse in the number of importers and imported HTS-10 products, followed by a slow recovery.

Economic message: The crisis sharply reduces the breadth of import participation, but the later decomposition shows that counts alone overstate the value-weighted importance of country-level entry and exit.

9.4 Figure 4: Various Extensive Margin Definitions by Quarterly and Annual Data and Table 3: Intensive and Extensive Margins, 2000–2002

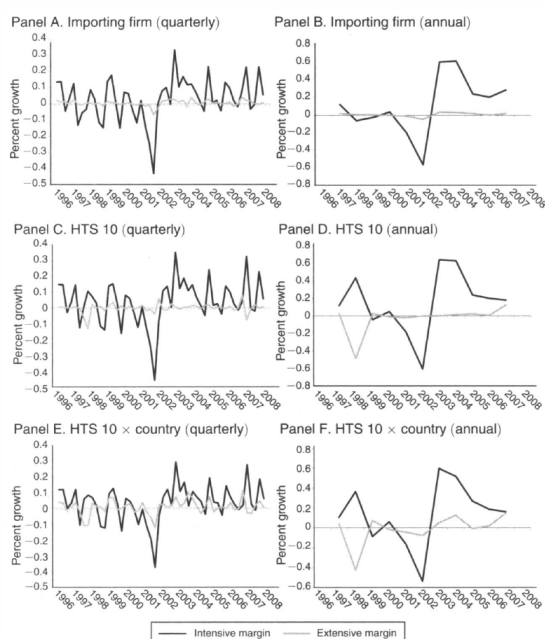


TABLE 3—INTENSIVE AND EXTENSIVE MARGINS, 2000–2002

	Percent total	Percent intensive	Percent extensive
Firm	–69	0.89	0.11
HTS 6	–69	1.00	0.00
HTS 10	–69	0.92	0.08
HTS 6 × Cty	–69	0.91	0.09
HTS 10 × Cty	–69	0.79	0.21

Figure 4: Extensive-margin definitions using quarterly and annual data.

Table 3: Intensive and extensive margins, 2000–2002.

Read: Figure 4 shows that the intensive margin explains most aggregate import growth and decline across several definitions. Table 3 quantifies this for 2000–2002: the extensive margin is small except under the finest HTS-10-by-country definition.

Economic message: Country-level extensive margins are not where most value adjustment occurs. This pushes the analysis toward within-firm margins.

9.5 Figure 5: Sub-Extensive Margin and Table 4: Sub-Intensive, Sub-Extensive, and Extensive Margins, 2000–2002

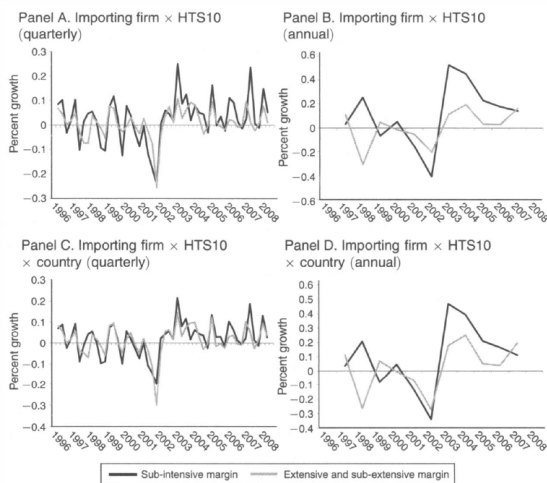


FIGURE 5. SUB-EXTENSIVE MARGIN

Figure 5: Sub-extensive margin.

TABLE 4—SUB-INTENSIVE, SUB-EXTENSIVE, AND EXTENSIVE MARGINS, 2000–2002

	Percent total	Percent sub-intensive	Percent sub-extensive	Percent extensive
HTS 6	–69	0.71	0.18	0.11
HTS 10	–69	0.56	0.33	0.11
HTS 6 × Cty	–69	0.54	0.35	0.11
HTS 10 × Cty	–69	0.44	0.45	0.11

Table 4: Sub-intensive, sub-extensive, and extensive margins.

Read: Figure 5 shows that within-firm product churning contributes meaningfully to import changes. Table 4 shows that the sub-extensive margin explains as much as 45 percent of the 2000–2002 import decline.

Economic message: The extensive margin becomes economically large once it is measured at the firm-input level. This is the empirical fact that motivates fixed costs of imported varieties in the model.

9.6 Figure 6: Different Firms Drop Different Products and Figure 7: Adjustment by Importer Size, 2000–2002

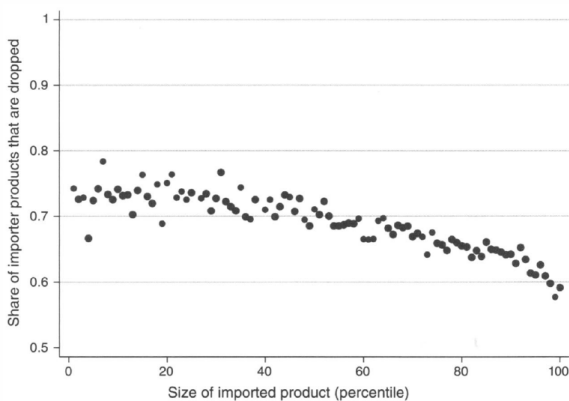


FIGURE 6. DIFFERENT FIRMS DROP DIFFERENT PRODUCTS

Figure 6: Different firms drop different products.

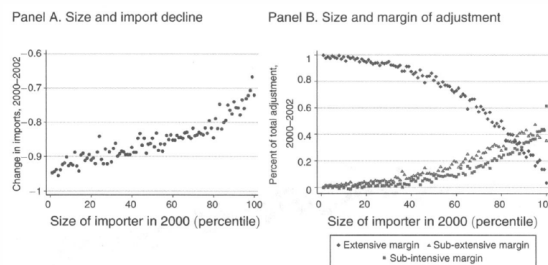


FIGURE 7. ADJUSTMENT BY IMPORTER SIZE, 2000–2002

Figure 7: Adjustment by importer size, 2000–2002.

Read: Figure 6 shows that many products are dropped by most importers but not all importers, so they remain in aggregate import data. Figure 7 shows that smaller importers experience larger import declines and rely more on extensive or sub-extensive adjustment.

Economic message: Perfect synchronization across firms is required for a firm-level dropped input to become a country-level dropped product. Size heterogeneity is therefore central for aggregate productivity effects.

9.7 Table 5: Impact of Product Extensive Margin on Imported Input Costs, 2000–2002 and Figure 8: Inventory Intensity and Import Decline

TABLE 5—IMPACT OF PRODUCT EXTENSIVE MARGIN ON IMPORTED INPUT COSTS, 2000–2002				
$\theta = 0.75$	$\mathcal{F}$	Weighted average of $\mathcal{F}_i$		
Percentiles included		All	(5,95)	(20,80)
HTS 6	1.000	1.087	1.046	1.034
HTS 10	0.992	1.110	1.068	1.060
HTS 6 $\times$ country	1.012	1.163	1.099	1.063
HTS 10 $\times$ country	1.004	1.176	1.096	1.097
Simple average	1.002	1.134	1.077	1.064

Table 5: Product extensive margin and imported input costs.

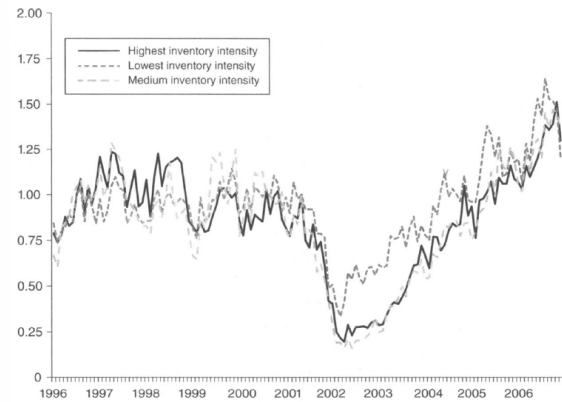


FIGURE 8. INVENTORY INTENSITY AND IMPORT DECLINE

Figure 8: Inventory intensity and import decline.

Read: Table 5 shows that aggregate data imply nearly no variety correction, while firm-level data imply sizable import bundle cost increases. Figure 8 addresses inventories by showing that imports in inventory-intensive sectors decline differently during the crisis.

Economic message: Dropped varieties raise true input costs, but this is only visible with firm-level data. Inventory behavior can attenuate or delay the production effect but does not eliminate the mechanism.

9.8 Figure 9: Export Varieties Did Not Decline Along with Import Varieties and Table 6: Baseline Simulation Parameters

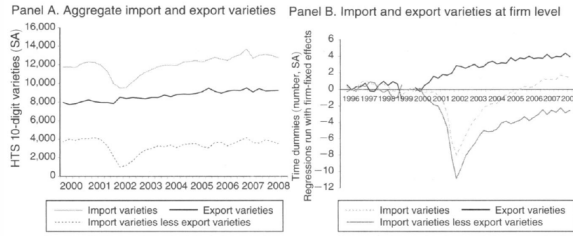


FIGURE 9. EXPORT VARIETIES DID NOT DECLINE ALONG WITH IMPORT VARIETIES

Figure 9: Export varieties did not decline along with import varieties.

θ	ρ	b	μ	α	λ	f	η	
0.75	0.75	1	2/3	1/3	2	0.0075	0.8	
w	r	C	P_N	ω	p_m^{pre}	$\hat{p}_m$	γ^{pre}	γ^{post}
50	50	1×10^8	0.96	0.2	1.70	1.15	0.83	0.89

Table 6: Baseline simulation parameters.

Read: Figure 9 shows that export varieties did not collapse like import varieties, which argues against a pure final-variety reduction story. Table 6 reports the benchmark parameters for the simulation, including the elasticities, input share, and pre/post domestic input shares.

Economic message: The fall in imported input varieties appears to be a production-input adjustment, not merely a reflection of firms producing fewer final varieties. The calibration then uses this evidence to discipline the model.

9.9 Figure 10: Concentration of Imports in the Data (2000) and in the Simulation and Figure 11: Adjustment and Size in Simulation

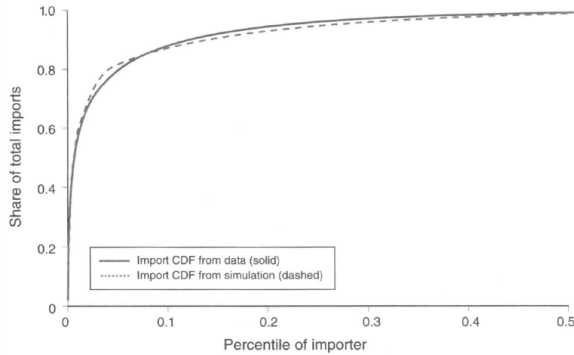


FIGURE 10. CONCENTRATION OF IMPORTS IN THE DATA (2000) AND IN THE SIMULATION

Figure 10: Concentration of imports in data and simulation.

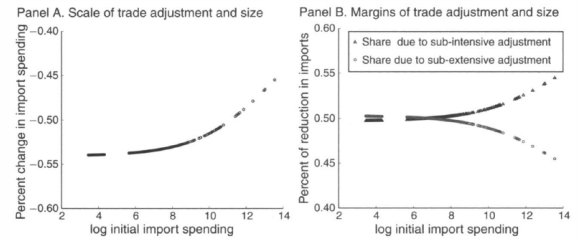


FIGURE 11. ADJUSTMENT AND SIZE IN SIMULATION

Figure 11: Adjustment and size in simulation.

Read: Figure 10 shows that the simulated distribution of import concentration closely matches the Argentine data. Figure 11 shows that the simulated model reproduces the empirical size pattern: larger firms adjust less and rely more on sub-intensive adjustment.

Economic message: The model is not only a theoretical exercise. It is calibrated to reproduce trade concentration and the size-gradient of adjustment, which are essential for aggregation.

9.10 Figure 12: Changes in Domestic Input Share and Productivity and Table 7: Simulation Results: Productivity

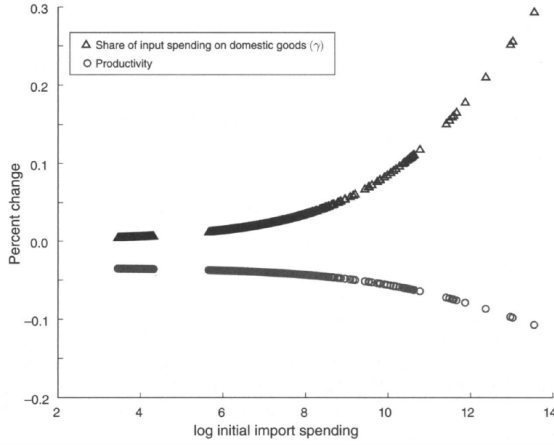


FIGURE 12. CHANGES IN DOMESTIC INPUT SHARE AND PRODUCTIVITY

Figure 12: Changes in domestic input share and productivity.

TABLE 7—SIMULATION RESULTS: PRODUCTIVITY

		$\Delta \ln PR$	$\ln \mathcal{F}$	$\Delta \ln \widehat{PR}$
(1)	Benchmark	-0.055	0.089	-0.066
(2)	No fixed costs	-0.043	0.000	-0.043
(3)	No roundabout production	-0.026	0.119	-0.038
(4)	$\rho = 0.50$	-0.068	0.037	-0.073
(5)	$\theta = 0.90$	-0.017	0.003	-0.017
(6)	$\rho = 0.50, \theta = 0.90$	-0.028	0.0025	-0.028
(7)	Adjusting for inventories, $\gamma^{post} = 0.86$	-0.024	0.034	-0.029
(8)	No capital goods, $\gamma^{post} = 0.87$	-0.033	0.049	-0.040

Table 7: Simulation results for productivity.

Read: Figure 12 shows that larger importers experience bigger increases in domestic input shares and larger productivity losses. Table 7 shows a 5.5 percent productivity decline in the benchmark and a 6.6 percent measured decline after accounting for mismeasurement.

Economic message: The sub-extensive margin materially changes productivity estimates. Ignoring fixed costs and dropped varieties understates the productivity loss.

9.11 Figure 13: Time-Series Paths of Productivity, Dropped Varieties, and Import Shares

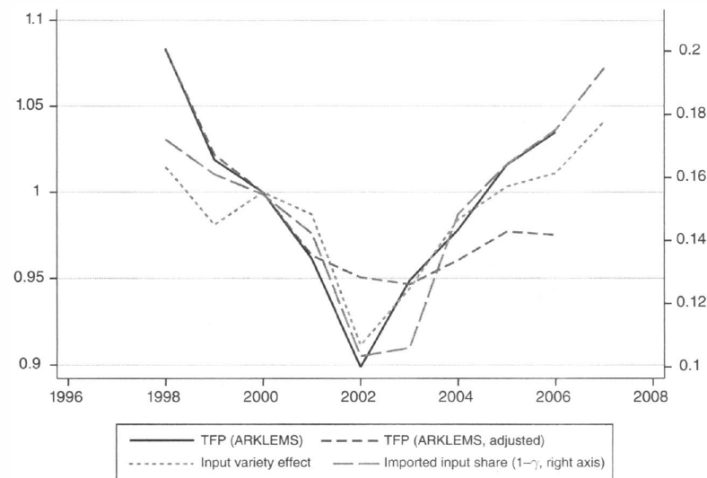


FIGURE 13. TIME-SERIES PATHS OF PRODUCTIVITY, DROPPED VARIETIES, IMPORT SHARES

Figure 13: Time-series paths of productivity, dropped varieties, and import shares.

Read: Figure 13 plots TFP measures, the imported input share, and the input variety effect over time. These variables decline around the crisis and recover afterward.

Economic message: The timing is consistent with the paper's mechanism: imported input use and variety losses move together with productivity through the crisis and recovery.

10 Short summary

- Argentina’s imports fell by 69 percent from 2000 to 2002, but firm entry/exit and country-level product entry/exit explain little of the value decline.
- Continuing importers changed the set of input varieties they imported. This within-firm product churning is the sub-extensive margin.
- The sub-extensive margin explains up to 45 percent of the import collapse.
- Small importers rely more on exit and sub-extensive adjustment; large importers mainly reduce quantities of continuing imports.
- Firm-level dropped varieties raise the true unit cost of imported input bundles by 6–13 percent under a CES assumption; aggregate data miss this.
- The model features heterogeneous firms, fixed costs of importing input varieties, imperfect substitution, markups, and roundabout production.
- Markups make scale reductions first-order relevant for welfare-relevant productivity.
- Roundabout production amplifies import-cost shocks because firms’ higher prices become other firms’ input costs.
- In the benchmark simulation, productivity falls by 5.5 percent, and measured productivity falls by 6.6 percent once missing varieties are accounted for.
- A model without fixed costs and sub-extensive adjustment produces a smaller productivity decline.

11 Conclusion

Gopinath and Neiman show that understanding trade adjustment in crises requires firm-level data. Aggregate data say that country-level extensive margins are small; the microdata reveal large within-firm input churning. This difference matters because imported intermediates enter firms’ production functions, and losing imported input varieties raises unit costs.

The theoretical contribution is to connect this micro adjustment to productivity. In their model, imported-input shocks reduce productivity through several channels: higher input costs, substitution toward domestic inputs, reductions in scale under markups, roundabout production, and mismeasurement from missing varieties. Fixed costs of importing make import demand non-homothetic, so the distribution of firm-level responses matters.

The paper’s main takeaway is that large import collapses are not merely expenditure reductions. They are changes in the production technology effectively available to firms. When firms lose access to imported input varieties, productivity can fall even without a decline in physical technology. This mechanism helps explain why measured TFP often collapses during large crises and why firm-level trade data are essential for understanding that collapse.